established even when there are broken connections.

Provider drivers are implementations that allow Octavia operators to choose the load balancing systems they want to use in their Octavia deployment. The following is a list of available provider drivers.

- A10 Networks OpenStack Octavia driver
- Amphora
- AmphoraV2
- F5 Networks provider driver for OpenStack Octavia by SAP SE
- OVN Octavia provider driver
- Radware provider driver for OpenStack Octavia
- VMware NSX

Amphora is used by Octavia (load balancing solution) as a default provider driver. OVN is chosen because, unlike Amphora, it does not require a VM to deploy which saves a lot of time and resources. But OVN has the following disadvantages:

- Layer-7 based load balancing is not possible because it supports TCP and UDP.
- It only supports the SOURCE_IP load balancing algorithm.
- Octavia flavours are not supported.
- Mixed IPv4 and IPv6 members are not supported.
- There are no health monitors in OVN's provider driver [*https://docs.openstack.org/ovn-octavia-provider/latest/admin/driver.html*].

## The setup

The following system configuration was used in this experiment.

- Operating system: Ubuntu 20.04 LTS (Server)
- Number of cores: 2
- RAM: 5GB
- Kernel version: 5.4.0

The following OpenStack services were used in this setup.

***Nova:*** This is the compute service used for providing virtual machines and bare metal servers.

***Neutron:*** This is used for providing Network-as-a-Service (NaaS) functionality between different interfaces.

***Octavia:*** This is a load balancing solution used in OpenStack.

***Keystone:*** Keystone is used for authentication and authorisation of users and service discovery.

***Placement:*** Placement provides HTTP API to allow access to resource usages which can help services manage their resources effectively.

***Glance:*** Glance is used to discover, register and retrieve virtual images.

## Installation and configuration

**OpenStack with Octavia:** OpenStack is installed in the server version of Ubuntu 20.04 LTS. To install it with Octavia, DevStack is used. Here, Octavia is installed as a DevStack plugin. We create a new user stack before installing OpenStack.

Octavia.create_and_delete_loadbalancers (155.007s)

*Create a loadbalancer per each subnet and then delete loadbalancer*

Overview | Details | Failures | Input task

Load duration: **82.488 s** Full duration: **155.007 s** Iterations: **5** Failures: **5** Started at: **2020-11-20T07:17:50**

Service-level agreement

| Criterion | Detail | Success |
|---|---|---|
| failure_rate | Failure rate criteria 0.00% <= 100.00% <= 0.00% - Failed | False |

Total durations

| Action | Min (sec) | Median (sec) | 90%ile (sec) | 95%ile (sec) | Max (sec) | Avg (sec) | Success | Count |
|---|---|---|---|---|---|---|---|---|
| octavia.load_balancer_create | 1.986 | 2.406 | 5.42 | 5.704 | 5.987 | 3.415 | 100.0% | 5 |
| octavia.wait_for_loadbalancers | 6.205 | 6.239 | 60.779 | 61.062 | 61.344 | 27.992 | 0.0% | 5 |
| *-> octavia.load_balancer_show (x29)* | *3.039* | *3.546* | *3.952* | *4.003* | *4.053* | *3.546* | *100.0%* | *2* |
| *-> octavia.load_balancer_show (x4)* | *0.198* | *0.235* | *0.236* | *0.236* | *0.236* | *0.223* | *100.0%* | *3* |
| **total** | **8.191** | **8.644** | **65.917** | **65.918** | **65.919** | **31.406** | **0.0%** | **5** |
| *-> duration* | *8.191* | *8.644* | *65.917* | *65.918* | *65.919* | *31.406* | *0.0%* | *5* |
| *-> idle_duration* | *0* | *0* | *0* | *0* | *0* | *0* | *0.0%* | *5* |

Figure 1: Result of *Octavia.create_and_delete_loadbalancers* task using Amphora

Octavia.create_and_delete_loadbalancers (54.822s)

*Create a loadbalancer per each subnet and then delete loadbalancer*

Overview | Details | Input task

Load duration: **15.243 s** Full duration: **54.822 s** Iterations: **5** Failures: **0** Started at: **2020-11-18T07:02:30**

Service-level agreement

| Criterion | Detail | Success |
|---|---|---|
| failure_rate | Failure rate criteria 0.00% <= 0.00% <= 0.00% - Passed | True |

Total durations

| Action | Min (sec) | Median (sec) | 90%ile (sec) | 95%ile (sec) | Max (sec) | Avg (sec) | Success | Count |
|---|---|---|---|---|---|---|---|---|
| octavia.load_balancer_create | 1.751 | 3.965 | 5.361 | 5.629 | 5.896 | 3.729 | 100.0% | 5 |
| octavia.wait_for_loadbalancers | 0.034 | 0.042 | 0.074 | 0.082 | 0.09 | 0.052 | 100.0% | 5 |
| *-> octavia.load_balancer_show* | *0.033* | *0.042* | *0.074* | *0.082* | *0.09* | *0.051* | *100.0%* | *5* |
| octavia.load_balancer_delete | 1.056 | 1.398 | 3.324 | 3.868 | 4.412 | 1.942 | 100.0% | 5 |
| **total** | **3.563** | **5.752** | **7.089** | **7.384** | **7.679** | **5.723** | **100.0%** | **5** |
| *-> duration* | *3.563* | *5.752* | *7.089* | *7.384* | *7.679* | *5.723* | *100.0%* | *5* |
| *-> idle_duration* | *0* | *0* | *0* | *0* | *0* | *0* | *100.0%* | *5* |

Figure 2: Result of *Octavia.create_and_delete_loadbalancers* task using OVN